作业12.

9.26 (1)

原式 $=\int_0^2 r^3 dr \int_0^{2\pi} d\theta \int_{-1}^{2} dz = 24\pi.$

9.27 (1).

原式 $=\int_0^a r^2 e^r dr \int_0^{\pi} \sin\phi \, d\phi \int_0^{2\pi} d\varphi$

$=4\pi e^2(a^2-2a+2)+8\pi.$

(4). 令 $0\le \rho\le 1$, $0\le\varphi\le\frac{\pi}{2}$, $0\le\theta\le 2\pi$

∴原式

$=\int_0^1 \frac{\rho^3 \ln(1+\rho^2)}{1+\rho^2} d\rho \int_0^{\frac{\pi}{2}} \sin\varphi\cos\varphi \, d\varphi \int_0^{2\pi} d\theta$

$=\frac{\pi}{4}(4\ln 2-2-(\ln 2)^2).$

28.(1). 柱坐标下,

$0\le r\le 2\sqrt{2}$; $0\le\theta\le 2\pi$; $0\le z\le r\sqrt{1+\sin^2\theta}$

∴原式 $=\int_0^{2\sqrt{2}} r\,dr \int_0^{2\pi} d\theta \int_0^{r\sqrt{1+\sin^2\theta}} 2z\,dz$

$=\int_0^{2\sqrt{2}} r^3 dr \int_0^{2\pi}(1+\sin^2\theta)\,d\theta = 48\pi.$

(3). 原式: $0\le r\le\sqrt{\cos 2\theta}$, $-\frac{\pi}{4}\le\theta\le\frac{\pi}{4}$ and $\frac{3}{4}\pi\le\theta\le\frac{5}{4}\pi$

$0\le z\le\frac{1}{2}r^2$

∴原式 $=2\int_0^{\sqrt{\cos 2\theta}} r\,dr \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} d\theta \int_0^{\frac{1}{2}r^2} dz$

$$= \frac{1}{24}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos^3 2\theta \, d\theta = \frac{1}{36}.$$

29. (1) 柱坐标：区域 $0 \le r \le 1$，$0 \le \theta \le \pi$

$r \le z \le \sqrt{r}$，被积函数 $f = r^2 \sin\theta \cos\theta z$.

$$\therefore \text{原式} = \int_0^1 r^3 dr \int_0^{\frac{\pi}{2}} \sin\theta\cos\theta \, d\theta \int_r^{\sqrt{r}} z \, dz$$

$$= \frac{1}{4}\int_0^1 (r^5 - r^7)$$

$$= \frac{1}{96}$$

(4) 球坐标：$0 \le \rho \le 3\sqrt{2}$，$0 \le \phi \le \frac{\pi}{4}$，$0 \le \theta \le \frac{\pi}{2}$.

$f = \rho^2$

$$\therefore \text{原式} = \int_0^{3\sqrt{2}} \rho^4 d\rho \int_0^{\frac{\pi}{4}} \sin\phi \, d\phi \int_0^{\frac{\pi}{2}} d\theta.$$

$$= \frac{486\pi}{5}(\sqrt{2}-1).$$

30. (2) ~~球坐标：$0 \le \rho \le 3\sqrt{2}$，$0 \le \phi \le \frac{\pi}{4}$，$0 \le \theta \le 2\pi$.~~

~~$V = \int_0^{3\sqrt{2}} \rho^2 d\rho \int_0^{\frac{\pi}{4}} \sin\phi \, d\phi \int_0^{2\pi} d\theta$~~

~~$=$~~

30.(2). 采用：柱坐标.

区域：$0 \le r \le 3$；$0 \le \theta \le 2\pi$，$r^2 \le z \le 18 - r^2$.

$\therefore V = \int_0^3 r\,dr \int_0^{2\pi} d\theta \int_{r^2}^{18-r^2} dz$

$= \int_0^3 2\pi r(18 - 2r^2)\,dr = 81\pi$.

(4) 柱坐标下：区域：$0 \le r \le \sqrt{z^2 + R^2}$，$0 \le \theta \le 2\pi$，$0 \le z \le H$.

$\therefore V = \int_0^H dz \int_0^{2\pi} d\theta \int_0^{\sqrt{z^2+R^2}} r\,dr$

$= \pi \int_0^H (z^2 + R^2)\,dz = \pi\left(\frac{1}{3}H^3 + R^2 H\right)$.

32. (1). $z = 6 - 3x - \frac{3}{2}y$

$\therefore$ xOy 平面投影：$0 \le x \le 2$，$0 \le y \le 4 - 2x$

$\therefore S = \iint \sqrt{1 + z_x^2 + z_y^2}\,dA$

$= \frac{7}{2} \times 4 = 14$

(2) 旋转面的方程为 $\vec{r} = (u, \sin u \cos\theta, \sin u \sin\theta)$

$\therefore \frac{d\vec{r}}{du} = (1, \cos u \cos\theta, \cos u \sin\theta)$

$\therefore \frac{d\vec{r}}{d\theta} = (0, -\sin u \sin\theta, \sin u \cos\theta)$

$\therefore S = \int_0^{\pi} du \int_0^{2\pi} |\vec{r}_u \times \vec{r}_\theta|\,d\theta = 2\sqrt{2}\pi + 2\pi \ln(1 + \sqrt{2})$.

(5). $z_x=-2x$, $z_y=2y$.

$\therefore S=\iint_D \sqrt{1+4x^2+4y^2}\,dA$,

$=\int_1^2 r\sqrt{1+4r^2}\,dr\int_0^{2\pi}d\theta$.

$=\frac{\pi}{6}(17\sqrt{17}-5\sqrt{5})$

8). 圆柱为 $x^2+y^2=R^2$ 和 $y^2+z^2=R^2$.

对圆柱1: $r=R$, $-\sqrt{R^2-y^2}\le z\le\sqrt{R^2-y^2}$.

即: $\vec{r}=(R\cos\theta, R\sin\theta, z)$

$\vec{r}_\theta=(-R\sin\theta, R\cos\theta, 0)$,

$\vec{r}_z=(0, 0, 1)$, $0\le\theta\le2\pi$, $-R|\cos\theta|\le z\le R|\cos\theta|$

$\therefore dA=|\vec{r}_\theta\times\vec{r}_z|\,d\theta dz=R\,d\theta dz$.

$\therefore S_1=R\int_0^{2\pi}d\theta\int_{-R|\cos\theta|}^{R|\cos\theta|}dz=8R^2$

$\therefore S=2S_1=16R^2$.

33.(2). $m=\int_0^1dx\int_{x^2}^1 xy\,dy=\frac{1}{6}$

质心 $\bar{x}=\frac{1}{m}\int_0^1dx\int_{x^2}^1 x^2y\,dy=\frac{4}{7}$

$\bar{y}=\frac{1}{m}\int_0^1dx\int_{x^2}^1 xy^2\,dy=\frac{3}{4}$

$\therefore$ 质心坐标为 $(\frac{4}{7},\frac{3}{4})$.

(4). 令 $t=y-1$ $\therefore \mu = t+1+|t|$.

$$\therefore m=\int_{-1}^{0} 2\sqrt{1-t^2}\,dt+\int_0^1 (2t+1)2\sqrt{1-t^2}\,dt.$$

$$=\pi+\frac{4}{3}$$

$\bar{x}=0$

$$\bar{y}=\frac{1}{m}\left(\int_{-1}^{0}(t+1)2\sqrt{1-t^2}\,dt+\int_0^1 (t+1)(2t+1)2\sqrt{1-t^2}\,dt.\right.$$

$$=\frac{16+15\pi}{16+12\pi}$$

36. (2). $I=\int_0^h dy\int_{-\frac{a}{2}(1-\frac{y}{h})}^{\frac{a}{2}(1-\frac{y}{h})} x^2\,dx=\frac{1}{48}a^3h.$

(4) 球坐标下:

$0\le r\le\sqrt{2},\ 0\le\theta\le 2\pi,\ \frac{\pi}{4}\le\varphi\le\frac{3}{4}\pi,$

$\therefore\ x^2+y^2=r^2\sin^2\varphi$

$$\therefore I=\int_0^{\sqrt{2}} r^4\,dr\int_{\frac{\pi}{4}}^{\frac{3}{4}\pi}\sin^3\varphi\,d\varphi\int_0^{2\pi}d\theta=\frac{8}{3}\pi.$$

37. 圆柱体的转动惯量张量为

$$\tilde{I}=\begin{bmatrix}\frac{1}{12}m(3R^2+H^2) & 0 & 0\\ 0 & \frac{1}{12}m(3R^2+H^2) & 0\\ 0 & 0 & \frac{1}{2}mR^2\end{bmatrix}$$

$\therefore\ \frac{1}{12}m(3R^2+H^2)=\frac{1}{2}mR^2$

即: $H:R=\sqrt{3}:2$

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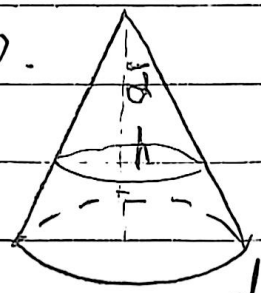
38.(1).



设距石英砂为x处的圆片上的一个小圆环dr，厚度为dx

产生引力：

$$dF = G\frac{dm}{(r^2+x^2)}\cdot\frac{x}{\sqrt{r^2+x^2}}$$

$$= 2\pi G\cdot\frac{x}{(r^2+x^2)^{\frac{3}{2}}}\,dr\,dx$$

记 $f = 2\pi G\frac{x}{(r^2+x^2)^{\frac{3}{2}}}$

$$\therefore F = \int_0^h dx\int_0^{x\tan\alpha} f\,dr = 2\pi kH(1-\cos\alpha)$$

(3) 设P位于r_0处，$r<R$.

∴位于$r>r_0$处的球壳对其内的引力为0

$$\therefore F = G\frac{M}{r_0^2} = \frac{G}{r_0^2}\cdot\frac{4}{3}\pi r_0^3 = \frac{4}{3}G\pi r_0$$

思考题:

设圆环中心轴为z轴

圆环管道截面圆心到z轴的距离为R

圆环截面半径为r.

截面上的角度为$\theta\in[0,2\pi)$; R绕z轴旋转的角度为$\varphi\in[0,2\pi)$.

∴环面上的一点$\vec{r}(x,y,z)$.

$$\begin{cases} x=(R+r\cos\theta)\cos\varphi \\ y=(R+r\cos\theta)\sin\varphi \\ z=r\sin\theta \end{cases}$$

$$\therefore \vec{r_\theta}=\frac{\partial\vec{r}}{\partial\theta}=(-r\cos\varphi\sin\theta,-r\sin\varphi\sin\theta,r\cos\theta).$$

$$\vec{r_\varphi}=\frac{\partial\vec{r}}{\partial\varphi}=(-(R+r\cos\theta)\sin\varphi,(R+r\cos\theta)\cos\varphi,0).$$

$$\therefore |\vec{r_\theta}\times\vec{r_\varphi}|=r(R+r\cos\theta).$$

$$\therefore S=\int_0^{2\pi}d\varphi\int_0^{2\pi}r(R+r\cos\theta)d\theta=4\pi^2Rr.$$

体积:柱坐标下截面圆方程$(\rho-R)^2+z^2\le r^2$.

$$\therefore R-r\le\rho\le R+r;\ -\sqrt{r^2-(\rho-R)^2}\le z\le\sqrt{r^2-(\rho-R)^2}$$

$$0\le\phi\le2\pi$$

$$\therefore V=\int_0^{2\pi}d\phi\int_{R-r}^{R+r}\rho d\rho\int_{-\sqrt{r^2-(\rho-R)^2}}^{\sqrt{r^2-(\rho-R)^2}}dz.$$

$$=4\pi\int_{R-r}^{R+r}\rho\sqrt{r^2-(\rho-R)^2}\,d\rho=4\pi\cdot(\tfrac{1}{2}\pi Rr^2)$$

$$=2\pi^2Rr^2$$

∴表面积为$4\pi^2Rr$. 体积为$2\pi^2Rr^2$